Key labeled features include the nose-mounted electro-optical and infrared (EO/IR) sensors, AI-driven flight control unit, autonomous navigation module, and concealed exhaust system to minimize thermal emissions. The figure also denotes the data-link and communication suite, enabling secure remote operation and coordination within a network-centric warfare (NCW) environment.

The Taranis demonstrator successfully performed autonomous taxi, takeoff, flight, and landing operations, proving its self-governing capabilities. As shown in the figure, its sleek, wing-integrated layout represents a balance between stealth and aerodynamic efficiency.

6.5.2.7 Boeing Phantom Ray – 2011 (Prototype)

The Boeing Phantom Ray, first unveiled in 2011, is an experimental stealth unmanned combat aerial vehicle (UCAV) prototype developed by Boeing Phantom Works (United States) as part of the company's advanced research into autonomous flight, stealth design, and networked combat systems. Derived from the X-45C Joint Unmanned Combat Air System (J-UCAS) program, the Phantom Ray embodies Boeing's early efforts to pioneer next-generation unmanned strike and reconnaissance aircraft with fully autonomous mission execution and low-observable (LO) design features (Masse, 2011).

The airframe of the Phantom Ray adopts a tailless flying-wing configuration, emphasizing stealth, endurance, and aerodynamic efficiency. Constructed with composite materials and coated with radar-absorbent coatings (RAM), it minimizes radar and infrared signatures. The aircraft measures approximately 11 meters in length, with a 15-meter wingspan, and has a maximum takeoff weight of around 16,000 kilograms. Its design prioritizes internal weapons storage to maintain stealth characteristics, allowing it to carry precision-guided bombs, electronic warfare payloads, or